Faculty of Science and Technology
Program Development in a Graphical Environment

Due: end of the class **Prof.: Nagat Drawel**

Instructions:

1. **Submit the full package:** Include all required project files.
2. I encourage you to use Git throughout your development process, make regular commits with descriptive commit messages and include the link to your Git repository as a comment at the top of your .java file.

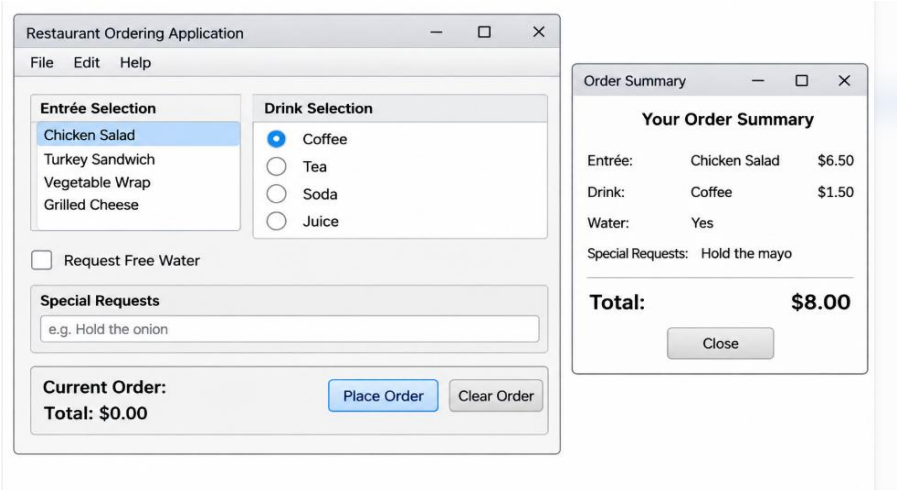
Problem Statement

The manager of the college campus restaurant wants to provide a computer terminal where students can place food and drink orders. Develop a JavaFX application that allows a student to select one entrée, one drink, request free water, and enter additional special requests. Students must be able to review their selections, see the total price, and clear or modify their order before placing it.

The application must be developed using the Model–View–Controller (MVC) architectural pattern.

1. Main Window: Order Selection

Create a user-friendly main window containing the following controls:



Order section	JavaFX control	Expected behaviour
Entrée Selection	ListView	Display the four entrée choices. The customer must select one entrée.
Drink Selection	RadioButton controls in a ToggleGroup	Display the four drink choices. Only one drink can be selected.
Special Requests	TextField or TextArea	Allow the customer to enter additional instructions
Water	CheckBox	Allow the customer to request free water. Water has no additional charge.
Current Order / Total	Label (or suitable display controls)	Display the current selections and the calculated total price.

Note: Water and additional special requests are two separate inputs. The water CheckBox must not be part of the Special Requests text.

2. Menu Bar and Menu Options

Add a MenuBar to the main window with the following menus and menu items.

File Menu

Place Order

- Open a separate order-summary window (a new JavaFX Stage).
  - Display the selected entrée, drink, and its price.
  - Display any additional special requests and whether water was requested.
  - Display the total order price.
- The customer must be able to close the summary window and return to the main window.
- Before displaying the summary, verify that the customer has selected one entrée and one drink. If either selection is missing, display an appropriate message and do not show the order summary.

Clear Order

- Clear every selection and input in the main window: entrée, drink, additional special requests, and water.
- Reset the total price to $0.00.
- The main window remains open.

Exit

- Close the application.

Edit Menu

These options clear only the specified part of the order. All other selections must remain unchanged.

- Entrée: Clear the selected entrée only.
- Drink: Clear the selected drink only.
- Special Request: Clear the additional special-requests text only. Do not clear the water CheckBox, entrée, or drink.

Help Menu

About: Display an information dialog containing the application name, version, and developer information. Use an Alert dialog for the About information.

3. Available Menu Items and Prices

Entrées: the customer must select one:

Entrée	Price
Chicken Salad	$6.50
Turkey Sandwich	$6.75
Vegetable Wrap	$6.00
Grilled Cheese	$5.00

Drinks: the customer must select one:

Drink	Price
Coffee	$1.50
Tea	$1.50
Soda	$2.00
Juice	$1.75

- Water — no additional charge.
- Additional special requests — no additional charge.

4. Application Architecture (MVC)

Your application must use the MVC architectural pattern and include the following classes.

A. Model – Order.java

- Represent the customer's order and store:
  - Selected entrée.
  - Selected drink.
  - Additional special requests.
  - Water selection.
- Store or provide access to the prices of the available entrées and drinks.
- Calculate and return the total price.
- The Model must not contain JavaFX GUI components.

B. View – OrderView.java

- Create and display the main JavaFX window and its controls.
- The View must not perform price calculations or contain business logic.

C. Controller – OrderController.java

- Handle user interactions with the GUI controls and menu items.
- Retrieve the customer's selections from the View.
- Update the Model when the customer's selections change.
- Request the total price from the Model and update the displayed total.
- Handle Place Order, Clear Order, Exit, About, and the individual Edit menu actions.
- Validate that one entrée and one drink have been selected before displaying the order summary.